

ABHISHEK YELMAMDI

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EDUCATION

PES University <i>B.Tech in Computer Science — CGPA: 8.42</i>	Bangalore, India 2023 – 2026
Appa Public School <i>SSLC — 70.45%</i>	Kalaburagi, India 2019

EXPERIENCE

Full Stack Developer <i>PES University — Smart Digital Library System</i>	Bangalore, India Jan 2026 – May 2026
- Built and owned the full-stack brainstorming room booking module — React frontend, Flask REST API, PostgreSQL — with JWT authentication and SRN-based misuse prevention. - Designed real-time library seat availability map using Socket.io integrated with a YOLOv11 CCTV-based occupancy detection pipeline. - Implemented chatbot with intent detection across 6 domains and multi-step conversation flows into the main web UI. - Implemented JWT authentication (HMAC-SHA256) with access tokens, refresh tokens, session management, and audit logging.	

PROJECTS

Transactional Inventory Reservation System | *Node.js, Express.js, PostgreSQL, REST APIs* | *Live*

- Implemented PostgreSQL row-level locking (`SELECT FOR UPDATE`) inside database transactions to prevent overselling during concurrent checkout requests.
- Designed reservation workflows (confirm, release, expiry) with explicit 409 Conflict and 410 Gone API responses.
- Developed concurrency test cases simulating simultaneous requests to validate transaction consistency.
- Built reservation workflows (confirm, release, expiry) with lazy expiry cleanup and one-active-booking-per-user constraint.

Stock Portfolio Risk Analysis Platform | *React, Flask, Python*

- Built a full-stack fintech platform with Monte Carlo simulations and ARIMA forecasting for portfolio risk prediction with interactive React dashboards.
- Integrated sentiment analysis from financial news APIs to generate market-driven insights.
- Built persistent portfolio history tracking with Chart.js visualizations and a React dashboard.

PUBLICATION

Data Driven Football Formation and Transfer Recommendation System

- Co-authored at the **2nd International Conference on AI and Robotics (AIR 2026)**, Nazarbayev University, Kazakhstan. Published under **Springer / IOES**.
- Built position-specific ML pipelines using XGBoost, K-Means clustering, and stacking ensembles — achieving 93.3% midfielder classification accuracy and $R^2=0.89$ for goalkeeper valuation.

TECHNICAL SKILLS

Languages: C++, Python, SQL
Full Stack: React.js, Node.js, Express.js, Flask, REST APIs
AI & ML: NLP, Time-Series Forecasting, Collaborative Filtering
Distributed Systems: Apache Kafka, Spark Streaming
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems
Tools: AWS, Docker, Git, Linux

CERTIFICATIONS

AWS Educate: Getting Started with Serverless (Lab Certificate)
AWS Badges — 3-day bootcamp: Logic Building, GitHub, Linux, Data Structures — credly.com/ABHISHEK-YELMAMDI